When Mutualism Becomes Costly: Abiotic Stress Shapes Grass Demography

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Abstract

Beneficial interactions between plants and fungi are ubiquitous and foundational to terrestrial ecosystems. Plant-fungal mutualisms are expected to strengthen under environmental stress, yet the relative importance of abiotic and biotic factors in eliciting mutualistic benefits remains unclear. We studied how the demographic effects of seed-transmitted fungal endophytes on native three grass host species varied in response to abiotic and biotic factors using geographically distributed field experiments along an aridity gradient, cross with herbivore access or exclusion. Vital rate models showed that all components of host fitness (survival, growth and fertility) declined with increasing, indicating reduced performance under abiotic stress. Declines were more pronounced in individuals hosting endophytes, suggesting that interactions that are beneficial under benign conditions can become costly under abiotic stress. On the other hand, herbivory had no detectable effect on host demography or on the demographic effects of hosting endophytes. Our results indicate that abiotic stress was the primary determinant of interaction outcomes between plants and fungal symbionts, but in the opposite direction as predicted by theory and prior work. These results highlight that symbiotic fungi can become liabilities under stress, with implications for the dynamics of plant-fungal mutualisms under global change.

Introduction

Plant-microbe symbioses are widespread and ecologically important. These interactions are famously context-dependent, where the direction and strength of the interaction outcome depend on the environment in which it occurs (Bronstein, 1994; Hoeksema and Bruna, 2015). For example, plant-microbe interactions that are beneficial under stressful conditions may become neutral or parasitic under benign conditions (Giauque et al., 2019). Variation in the nature of biotic and abiotic stress is a key driver of these context-dependent outcomes.

Microbial symbioses can substantially enhance plant resilience to environmental stress. Under biotic stress, such as herbivory, microbial symbiosis can benefit host plants by facilitating the production of secondary compounds that deter feeding or exert direct toxicity, thereby reducing herbivore growth, survival, and oviposition (Atala et al., 2022; Bastias et al., 2017; Vega, 2008). Similarly, under abiotic stress, such as low precipitation, nutrient limitation, or high temperature, symbiotic microbes can enhance host tolerance through a variety of mechanisms (Afkhami and Rudgers, 2008; Clay and Schardl, 2002). Symbionts including mycorrhizal fungi, rhizobia, and endophytic bacteria can improve water and nutrient acquisition by increasing root surface area, facilitating nutrient mobilization, or promoting nitrogen fixation, allowing plants to maintain growth and metabolic function under drought or poor soil conditions (Amijee et al., 1989). In addition, many symbiotic microbes stimulate the production of antioxidant enzymes, such as superoxide dismutase and catalase, which reduce oxidative damage caused by stressors like drought, salinity, and heat (Ruiz-lazona et al., 1996). By improving resource acquisition, modulating plant physiology, and enhancing stress-responsive metabolism, these symbiotic interactions help host plants buffer against environmental stochasticity and extreme conditions across a wide range of ecosystems (Fowler et al., 2023).

Although microbial symbioses often enhance plant performance, they can also impose substantial costs, particularly in resource-limited environments. Maintaining symbionts requires hosts to allocate valuable carbon and nutrients to sustain their microbial partners. For instance, mycorrhizal fungi may require up to 20% of photosynthetically fixed carbon (Soudzilovskaia et al., 2015; Thirkell et al., 2020), and under abiotic stress, this investment can outweigh the benefits of the symbiosis (Bronstein, 2001; Johnson et al., 1997), reducing host survival, growth, reproduction, or competitive ability (Klironomos, 2003). Similarly, systemic and vertically transmitted endophytes can shift from mutualistic to antagonistic interactions when they produce stromata that abort host inflorescences or impose energetic costs (Saikkonen et al., 2004). Experimental studies further show that under low nutrient or water conditions, endophyte-infected grasses may produce fewer tillers and lower total biomass compared to uninfected plants, reflecting the costs of sharing limited resources (Ahlholm et al., 2002).

Despite growing interest in the role of symbiosis in shaping host performance, few studies have tested under natural conditions how abiotic and biotic stressors jointly influence the demographic effects of symbionts along environmental gradients (Louthan et al., 2015). Most existing work has been conducted in controlled or common garden experiments, where, for example, herbivory but not drought reduced plant growth, and no interaction between the two stressors was observed (Emery et al., 2010). Studying the combined effects of abiotic and biotic stressors under natural conditions is particularly important in the context of global change, as shifting climate patterns and altered herbivore pressures may change the balance of costs and benefits in plant–microbe interactions, potentially influencing species distributions and ecosystem function. Here, we address this gap by examining the joint effects of biotic and abiotic stress on grass-endophyte symbioses in native populations.

Grass endophytes, particularly those that are vertically transmitted through seeds, provide an especially tractable system for studying the ecological dynamics of symbiosis, as their infection status can be experimentally manipulated and compared across hosts. One of the most well-studied systems involves cool-season grasses and *Epichloë* fungi. Previous studies have shown that endophyte infection frequency increases more under ambient herbivory than when mammalian and insect herbivory are experimentally reduced (**?**). Infection prevalence can also vary with environmental conditions, such as long-term fluctuations in precipitation and temperature (Fowler et al., 2025). Host responses to endophytes are highly variable, and even when endophytes do not directly enhance survival under stress, they can still improve host fitness by promoting growth and reproduction, thereby offsetting potential survival costs (Yule et al., 2013). Together, these characteristics make grass–*Epichloë* symbioses an excellent model for investigating how biotic and abiotic stressors shape context-dependent mutualistic interactions.

Working across an aridity gradient in the south-central US, we investigated how endophyte symbiosis influences host-plant demography under varying abiotic stress. This region provides an ideal system to study biologically realistic stress gradients because it spans a strong east–west precipitation gradient, with mesic conditions in the east transitioning to increasingly arid conditions in the west. At each site along the gradient we also tested herbivory as a biotic stressor to evaluate its potential interactive effects with abiotic conditions. Specifically, we studied the symbiotic association between three native cool-season grass species (*Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis*) and their vertically transmitted fungal symbionts (*Epichloë* spp.). Our experiment was designed to address the two questions : (i) Under combined abiotic stress and herbivory, do endophytes provide fitness benefits or impose costs on their host plants? (ii) Which factors (abiotic vs. biotic) influence whether the grass-endophyte interaction benefits or harms the host plant? We predicted that endophytes will enhance host performance under low precipitation by mitigating drought stress. These benefits are expected to be smaller in fenced plots with reduced herbivory, where the contribution of symbionts to host performance is less critical. We further predicted that the grass–endophyte symbiosis will be less mutualistic under high stress conditions, particularly when herbivores presence adds biotic stress. While both types of stress are expected to influence the outcome, we anticipated that abiotic stress will be the primary driver of mutualistic benefits, with biotic stress modulating the magnitude of those benefits.

Materials and methods

Study system

Our study focused on three cool-season perennial grasses native to woodland and prairie habitats of eastern North America: *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis* (subfamily Pooideae) (Shaw, 2011). *A. hyemalis* is a small, short-lived perennial that germinates from autumn to late winter and typically flowers between March and July. *E. virginicus* is a larger, long-lived species that begins flowering as early as March or April and continues throughout the summer. *P. autumnalis* has a similar growth form to *E. virginicus* and flowers from late spring through summer. All three species host *Epichloë* fungal endophytes, which are primarily transmitted vertically through seeds. *Agrostis hyemalis* is associated with *Epichloë amarillans* (White Jr, 1994), *E. virginicus* typically hosts *Epichloë elymi* (Liu, 2023), and *P. autumnalis* can also harbor endophyte taxa capable of horizontal transmission via sexual stromata on inflorescences or asexual conidia on leaf surfaces (Clay and Schardl, 2002; Gundel et al., 2020; Leuchtmann et al., 2014). Horizontal transmission is rare, influenced by environmental and genetic factors, and was not observed in our study system. To examine the effects of endophytes on host species across a stress gradient, we established eight plots (1.5 m $\times$ 1.5 m) at each of seven sites along an aridity gradient in Louisiana and Texas (Fig. 1A-D). Each plot contained 15 individuals of a single species, with half of the plots protected from herbivores using exclusion cages to assess the interactive effects of endophytes and herbivory (Fig. 1E). In summer–fall 2021, seeds were collected from naturally symbiotic source populations and either heat-treated to remove endophytes or left untreated, producing symbiont-free (E^-) and symbiotic (E^+) plants from the same genetic lineages.

Source populations and symbiont removal

Plants used in the common garden experiment were derived from natural (source) populations throughout the natural distribution of the species (Fig. 1,Table S-1). Seeds were collected from these source populations, and some were heat-treated at 60°C for approximately five days (120 hours) to produce endophyte-negative plants (E^-), a method that eliminates endophytes without affecting seed viability. All seeds, both heat-treated and non-heat-treated, were planted in the Rice University greenhouse, where seedlings were fertilized every two weeks. Seedlings were then vegetatively propagated to produce enough individuals for the experiment. Before field planting, we confirmed endophyte status (E^+ or E^-) of all seedlings using either microscopy or an immunoblot assay. Leaf tissues were stained with aniline blue lactic acid and viewed under a compound microscope at 200x–400x to identify fungal hyphae. The immunoblot assay (Phytoscreen field tiller endophyte detection kit, Agrinostics Ltd. Co.) uses monoclonal antibodies that target proteins of *Epichloë* spp. and a chromagen to visually indicate presence or absence. Both methods yield similar detection rates.

Common garden experimental design and climate data

We conducted our common experiment across spanning the precipitation gradient from Louisiana to central Texas, USA (Fig. 1, Table S-2). To characterize the abiotic stress gradient, we collected daily temperature and precipitation data from PRISM (Hart and Bell, 2015). These daily values were used to calculate the mean temperature (°C) and cumulative precipitation (mm) for each year, spanning from the time plants were transplanted to when demographic data were collected. During our study period (2023-2025) realized precipitation at these sites corresponded well to 30-year normals, with wetter eastern sites and drier western sites. Mean annual temperature did not differ significantly across sites (Fig. S-1), so we focus on precipitation as the primary abiotic factor that varied across sites. This gradient also overlaps and exceeds the natural distributions of our focal species, ensuring realistic exposure to climatic and biotic conditions (Fig. 1).

Plots were randomly assigned one of four initial endophyte frequency treatments (80%, 60%, 40%, or 20%) and one of two herbivore access treatments (fenced to exclude herbivores or unfenced to allow access). In the fenced plots, insecticides were applied periodically to eliminate small herbivorous insects, as *Agrostis hyemalis*, *Epichloë amarillans*, and *Poa autumnalis* are naturally subject to herbivory by both insects and larger mammals. To ensure that our herbivory treatment reflected the level of herbivory experienced by individual plants, we counted the number of damaged plants per plot and estimated the proportion of damaged plants per site and herbivore treatment. Our results confirmed that the herbivory treatment was effective (Fig. 1F).

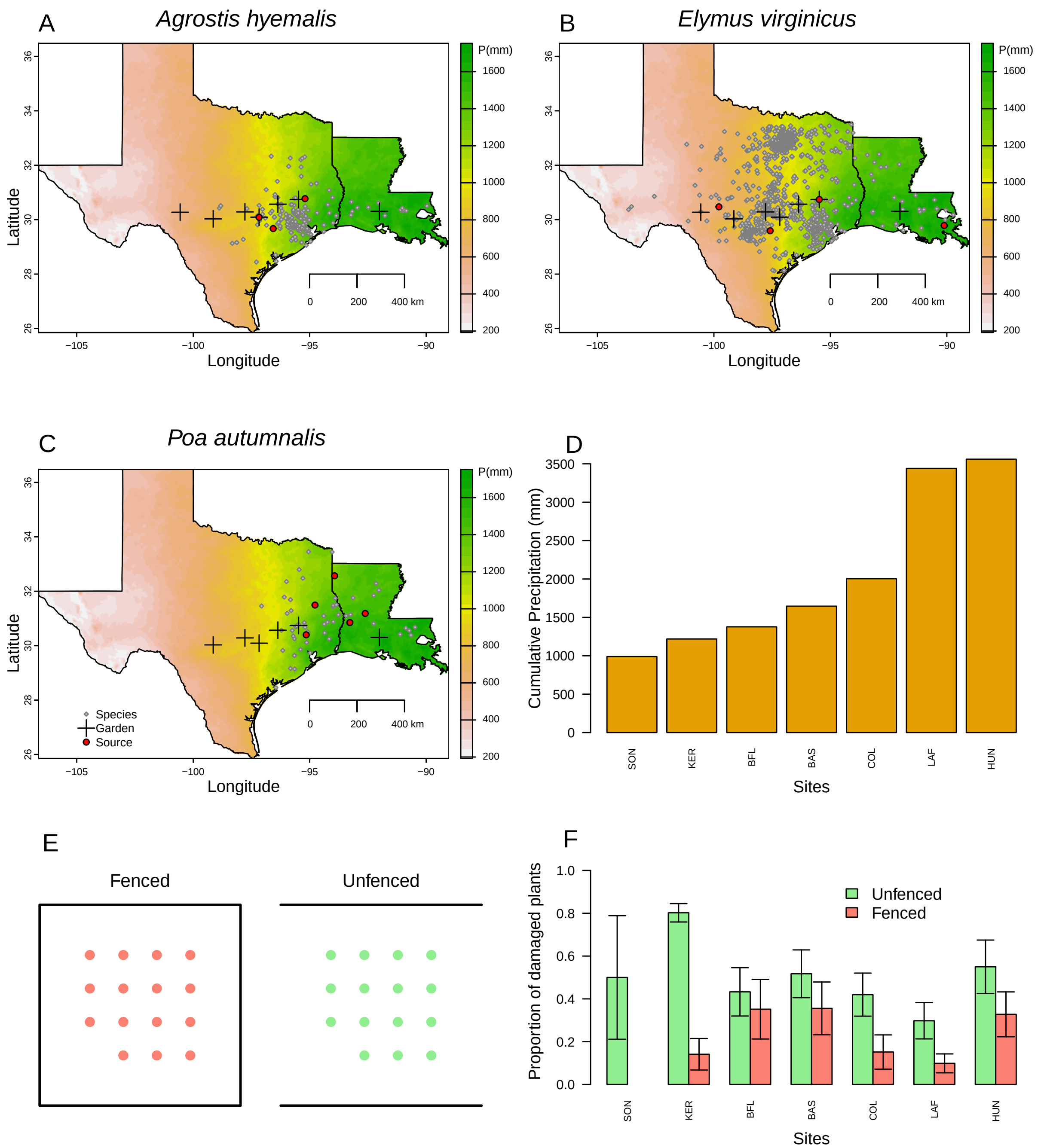

Figure 1: Distribution of common garden sites along a natural aridity gradient from Texas to Louisiana, US for (A) *Agrostis hyemalis*, (B) *Elymus virginicus*, and (C) *Poa autumnalis*. Climate data in A–C are based on 30-year normals of climate predictions (1993–2023) from PRISM data (Hart and Bell, 2015), which illustrate the precipitation gradient across sites (scale on the right is in mm. Red dots indicate the locations of source populations, while grey dots represent Global Biodiversity Information Facility (GBIF) occurrence records for each species within the study area (GBIF, 2025a,b,c). D) Cumulative precipitation during the study period (May 2023 – June 2025) at each site, ordered west to east. Values are also derived from PRISM climate data (Hart and Bell, 2015).

Demographic data collection

We collected demographic data, including survival, growth, and reproduction, during the flowering season of each species: *Agrostis hyemalis* and *Poa autumnalis* were censused in May, and *Elymus virginicus* was censused in June of 2023, 2024, and 2025. For each individual, plant survival was recorded as alive or dead and, for surviving plants, size was measured as the number of tillers. Growth was quantified as the log ratio of the number of tillers between consecutive censuses (i.e., $\log(\text{tiller}_{t+1}/\text{tiller}_t)$) for plants that were alive in both years. We recorded the number of inflorescences per plant for all three species and counted the number of spikelets on up to three inflorescences from reproducing individuals of *Poa autumnalis* and *Elymus virginicus*.

Assessing whether endophytes confer benefits or impose costs under abiotic and biotic stress

To assess whether endophytes confer benefits or impose costs under abiotic and biotic stress, we first developed two candidate models for each vital rate: one quadratic and one monotonic. All models shared the same hierarchical structure and were applied to survival, growth, and fertility. Survival was modeled with a Bernoulli distribution, growth with a Gaussian distribution, the number of inflorescences with a zero-inflated negative binomial distribution, and the number of spikelets with a negative binomial distribution. We modeled all vital rates (survival, growth, and fertility) using species-specific fixed effects for symbiosis (endophyte presence), precipitation, and herbivore access, including all relevant two- and three-way interactions. Each observation i corresponds to a plant in a given year; individuals can appear in multiple years. For simplicity, we omit the full subscript notation for species, plot, population, and site-year, which is explicitly included in the Stan model. We present the quadratic model here (Eq. 1) because it encompasses the linear model as a special case.

$$
\begin{aligned}
\mu_i = {} & \beta_0[Sp_i] + \beta_P[Sp_i]P_i + \beta_{P^2}[Sp_i]P_i^2 + \beta_{endo}[Sp_i]endo_i + \beta_{herb}[Sp_i]herb_i \\
& + \beta_{endo\times P}[Sp_i]endo_i P_i + \beta_{endo\times P^2}[Sp_i]endo_i P_i^2 + \beta_{herb\times P}[Sp_i]herb_i P_i \\
& + \beta_{herb\times endo}[Sp_i]herb_i endo_i + \beta_{endo\times herb\times P}[Sp_i]endo_i herb_i P_i \\
& + \phi_{\text{site}[i],Sp_i} + \omega_{\text{source}[i],Sp_i} + \rho_{\text{plot}[i],Sp_i}.
\end{aligned}
\tag{1}
$$

Here, P_i is total precipitation (mm) averaged over the 12 months preceding the census, $endo_i$ indicates endophyte presence, and $herb_i$ indicates herbivore access. Random effects account for background variation at multiple hierarchical levels: $\phi_{\text{site}[i],Sp_i}$ captures site-to-site heterogeneity not explained by precipitation, $\omega_{\text{source}[i],Sp_i}$ captures variation associated with the provenance of transplants, and $\rho_{\text{plot}[i],Sp_i}$ captures plot-to-plot variation. Each species has its own random effect for each site, source, and plot, while the variance parameters (σ_{site}, σ_{source}, σ_{plot}) are shared across species. This hierarchical structure allows the model to "borrow strength" across species, improving estimates of fixed effects and background variation, while still capturing species-specific responses to environmental heterogeneity. We confirmed that models effectively captured key aspects of the grass demography data, including means, standard deviations, and quantiles using a posterior predictive check (Fig. S-2).

Secondly, to determine the best-supported model for each vital rate, we compared candidate models (quadratic and monotonic) using leave-one-out cross-validation (LOOCV) (Vehtari et al., 2017). LOOCV combines validation and training by iteratively holding out one observation as a test set while fitting the model on the remaining $n-1$ observations. This process is repeated n times, once for each observation, resulting in n fitted models (Silva and Zanella, 2024). The overall test error estimate is then obtained by averaging the prediction errors across all n models (Eq. 2) for each vital rate.

$$CV_n = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{2}$$

Although we evaluated quadratic models, LOOCV showed that they provided minimal improvement over monotonic models (Table S-3). Therefore, all subsequent analyses used the monotonic model, which captures the observed patterns while maintaining model simplicity.

Finnaly, for each vital rate, precipitation was summarized at key quantiles (10th, 25th, 50th, 75th, and 90th percentiles) to represent the range of site-level abiotic stress. For each species (*Agrostis hyemalis*, *Elymus virginicus*, *Poa autumnalis*) and herbivory treatment (Fenced or Unfenced), we computed posterior predictions of survival probability for individuals with endophytes (E^+) and without endophytes (E^-) using the best Bayesian model outputs (Eq. 1). The difference ($\Delta = E^+ - E^-$) was calculated for each posterior sample to estimate median effects and 90% credible intervals. We also computed the posterior probability that $\Delta > 0$, representing the likelihood that endophytes confer benefits on vital rates under the given climate and herbivory conditions.

Drivers of endophyte effects on on cool-season grass demography

To determine the primary drivers of endophyte-mediated effects on grass performance, we focused on interactions with climate and herbivory to identify when endophytes confer benefits or costs under multiple stressors (abiotic and biotic). The endophyte $\times$ climate and endophyte $\times$ herbivory $\times$ climate interactions summarize how the benefits of endophyte presence on survival, growth, or reproduction change across precipitation gradients and herbivory treatments. The endophyte $\times$ herbivory interaction indicates whether endophytes mitigate or exacerbate herbivore impacts, and together these interactions highlight the environmental and biotic contexts that enhance or reduce symbiotic benefits.

Effect sizes were quantified using the posterior distributions from the hierarchical Bayesian models (Eq. 1). For survival, coefficients on the logit scale were exponentiated to obtain odds ratios (OR), representing the multiplicative change in the probability of survival associated with endophyte presence under different conditions. For flowering and spikelet production, coefficients were exponentiated to obtain incidence rate ratios (IRR), representing the proportional change in reproductive output. Growth was modeled with a Gaussian distribution, and effect sizes were expressed as the expected change in log-transformed growth per unit change in the predictor. For each species and vital rate, we summarized effects as mean OR, IRR, or growth change with 95% credible intervals. We also computed the posterior probability that OR or IRR exceeded 1, or that growth effects were positive, providing a direct measure of the likelihood that endophytes confer benefits under specific combinations of climate and herbivory. This approach allows comparison of the magnitude and certainty of abiotic, biotic, and interactive drivers on grass demography in biologically interpretable units.

All models were implemented in R (R Core Team, 2023) using Stan via the `rstan` interface (Stan Development Team, 2024).

Results

Endophyte effects on host fitness components across abiotic and biotic stress gradients

Across fitness components, *Epichloë* endophytes generally enhanced host performance under wetter conditions, while their effects were neutral or occasionally costly under drier conditions or in the presence of herbivores. Benefits were most apparent when herbivores were excluded.

For survival, endophytes effect was neutral or slightly costly at low precipitation, but became increasingly positive with wetter conditions, particularly in unfenced plots (Fig. 2; Fig. S-3). In *Agrostis hyemalis* and *Elymus virginicus*, median $\Delta(E^{+}-E^{-})$ ranged from -0.71 to 0.10 under dry conditions ($\Pr(\Delta>0)=0.37$–0.61; Table S-4) and increased to 0.95–1.54 with high precipitation in unfenced plots ($\Pr(\Delta>0)=0.69$–0.84;Table S-4). In fenced plots, positive effects emerged only at the highest precipitation and were weaker (Fig. S-3). In contrast, *Poa autumnalis* showed little benefit when herbivores were presented (median $\Delta=-0.71$ to 1.54, $\Pr(\Delta>0)=0.02$–0.08), but weakly positive survival when herbivores were excluded (median $\Delta=-0.13$ to 0.11, $\Pr(\Delta>0)=0.57$–0.71).

Epichloë endophytes enhanced growth with increasing precipitation across species, though magnitude and certainty varied with herbivory. In *A. hyemalis*, effects were neutral at low-to-moderate precipitation in unfenced plots (median $\Delta = 0.006$–0.093) and became more positive at high precipitation (median $\Delta = 0.170$–0.187; $\Pr(\Delta > 0) = 0.56$–0.74; Table S-5). Fenced plots showed similar but smaller effects. *E. virginicus* showed minor or negative effects under dry conditions (median $\Delta = -0.123$ to -0.008; $\Pr(\Delta > 0) = 0.35$–0.49), turning beneficial at higher precipitation, especially in unfenced plots (median $\Delta = 0.207$–0.238; $\Pr(\Delta > 0) = 0.81$–0.82). In *P. autumnalis*, effects were context-dependent: unfenced plots showed neutral to mild benefits at low precipitation (median $\Delta = 0.192$–0.258; $\Pr(\Delta > 0) = 0.69$–0.85), becoming strongly positive at high precipitation, while fencing amplified benefits across the gradient (median $\Delta = 0.461$–1.179; $\Pr(\Delta > 0) = 0.98$–1.00).

Epichloë endophytes also increased flowering and spikelet production, with effects generally stronger under wetter conditions (Figs. 4, 5; Figs. S-6). Flowering benefits were consistent across species, though decreased under dry conditions, with herbivore exclusion having little effect for *A. hyemalis* and *E. virginicus*, but enhancing benefits for *P. autumnalis* (median Δ up to 6.84; $\Pr(\Delta > 0) \sim 1.00$; Table S-6). Spikelet production was similarly context-dependent: *E. virginicus* and *P. autumnalis* showed negative or weak effects under low precipitation, shifting to positive effects under high precipitation, with fencing amplifying benefits across the gradient (Table S-7).

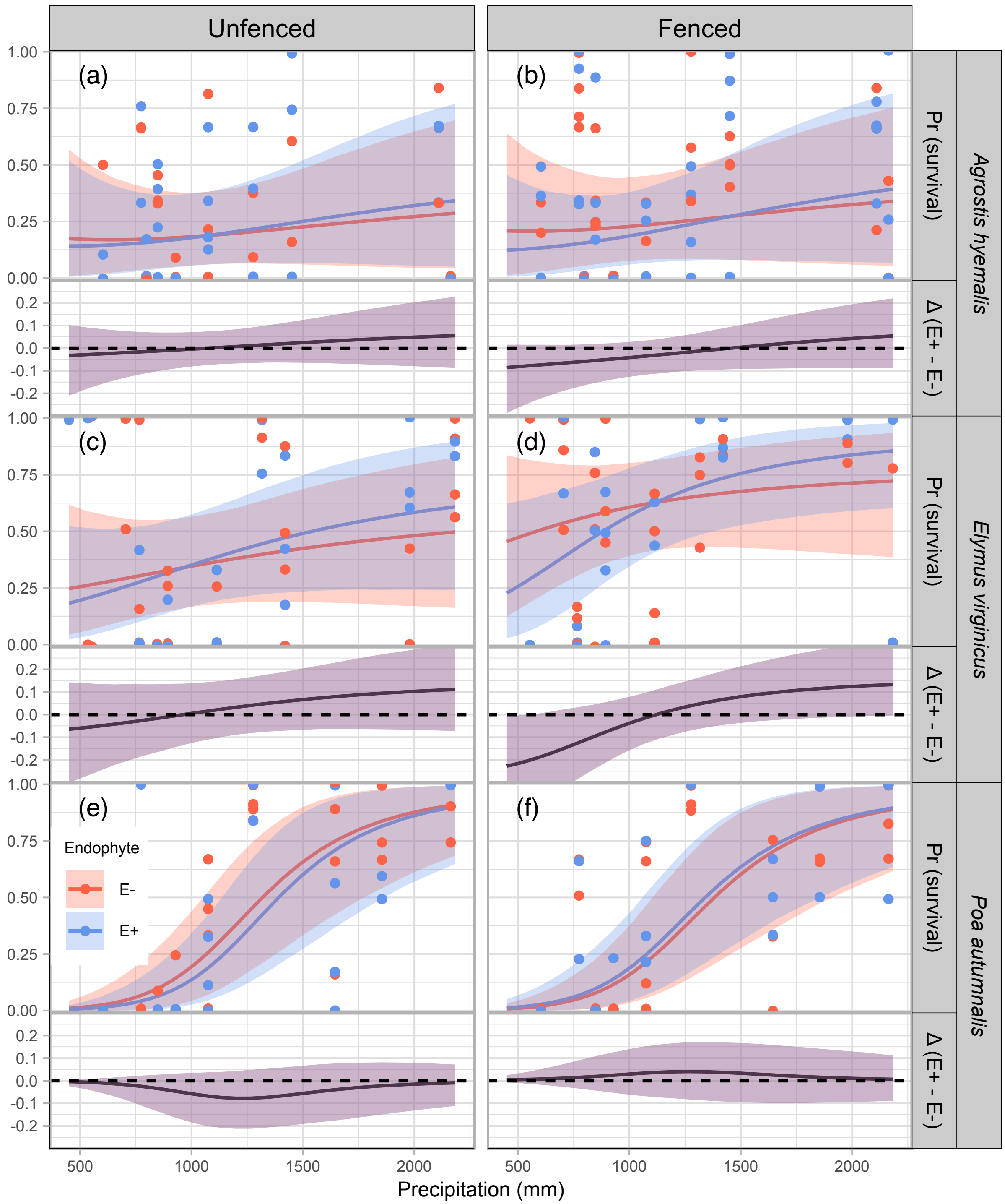

Figure 2: Predicted survival rate (top panels) and endophyte effect ($\Delta = \mathrm{E}^{+} - \mathrm{E}^{-}$, bottom panels) across precipitation gradients. Lines show posterior means for E^{+} (blue) and E^{-} (red), with shading indicating 90% credible intervals. Points represent mean observed survival per plot, jittered for visualization. The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

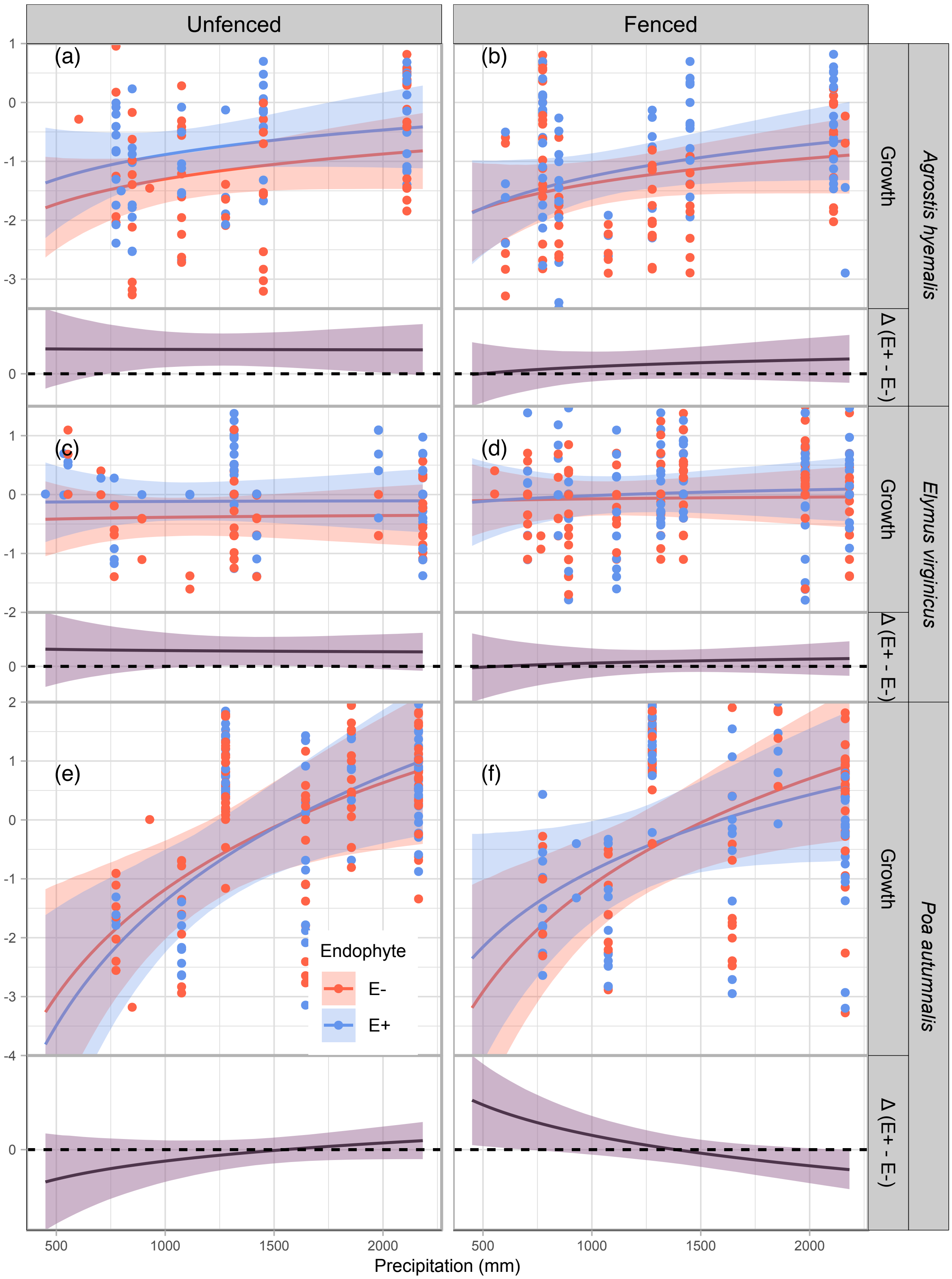

Figure 3: Predicted relative growth from year to the next year (top panels) and endophyte effect ($\Delta = E^+ - E^-$, bottom panels) across precipitation gradients. Lines show posterior means for E^+ (blue) and E^- (red), with shading indicating 90% credible intervals. Points represent mean observed growth per plot, jittered for visualization. The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

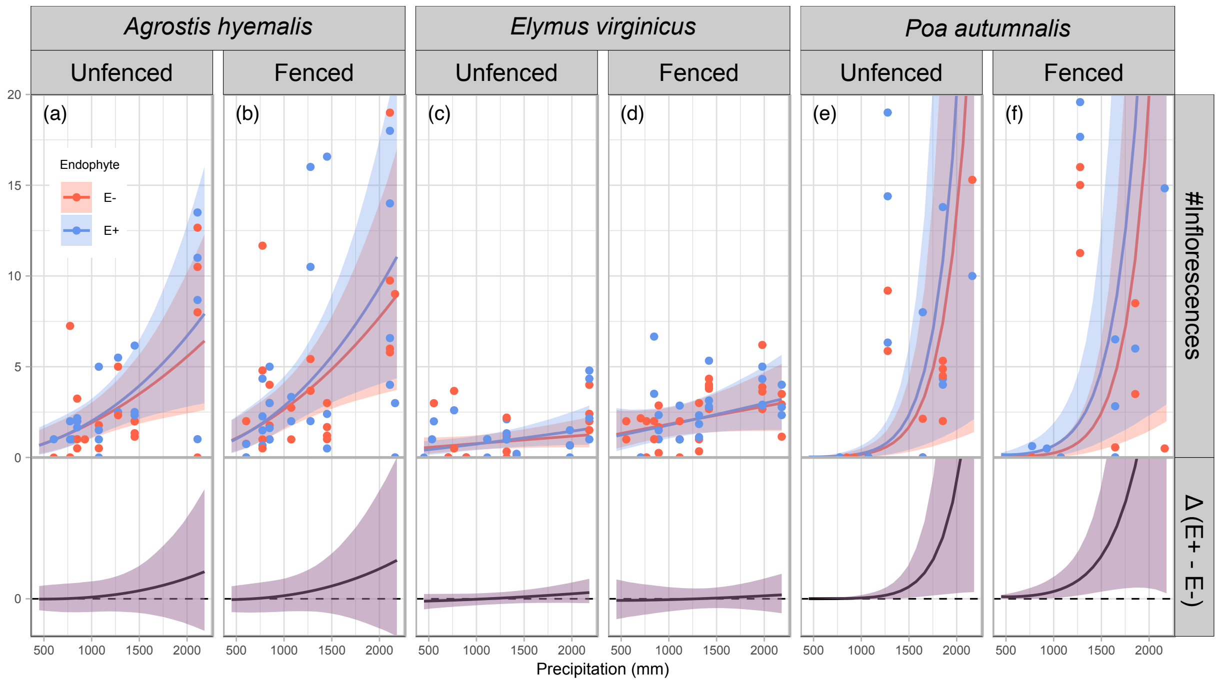

Figure 4: Predicted number of inflorescences (upper panels) and the difference between endophyte-infected (E^+) and endophyte-free (E^-) plants ($\Delta(E^+–E^-)$, lower panels) are shown across a gradient of precipitation. Shaded areas represent 90% credible intervals. Points indicate observed means,jittered for visualization. Panels are grouped by species and herbivory treatment (unfenced = with herbivory; fenced = herbivory excluded).

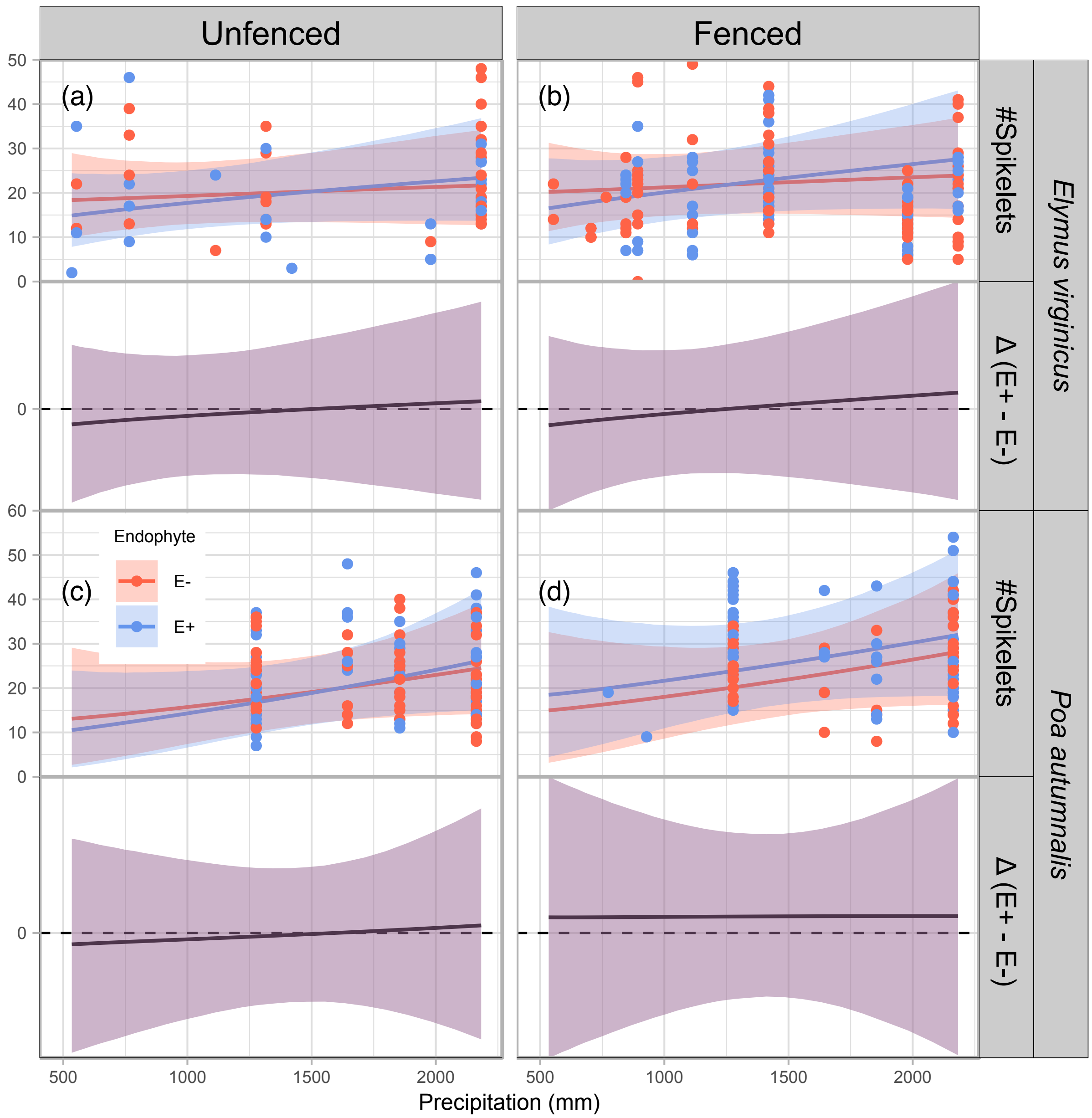

Figure 5: Predicted number of spikelets (upper panels) and endophyte effect ($\Delta = E^{+} - E^{-}$, bottom panels) across precipitation gradients. Lines show posterior means for E^{+} (blue) and E^{-} (red), with shading indicating 90% credible intervals. Points represent mean observed spikelets per plot, slightly jittered for visualization. Spikelets were measured for only two species (see Methods for details). The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

Abiotic factors as the primary drivers of fitness components across the stress gradient

Endophytes were most mutualistic under wet conditions or when hosts were exposed to herbivores, but they reduced growth or survival under dry conditions with low herbivory

(Fig. 6). Survival increased under wetter conditions due to the Endophyte × Climate interaction (OR = 1.21–1.36; Pr(OR $>$ 1) = 0.68–0.81;Table S-8). Herbivory modified these benefits in a species-specific manner: *Poa autumnalis* showed strongly enhanced survival when herbivores were present (OR = 1.94; Pr = 0.88;Table S-8), whereas responses in *Agrostis hyemalis* and *Elymus virginicus* were weak or uncertain. The three-way interaction further boosted survival in *A. hyemalis* and *E. virginicus* under herbivore exclusion in wet climates (OR = 1.15–1.45; Pr = 0.63–0.77;Table S-8), but had little effect on *P. autumnalis*. Growth responses were modest. Endophyte × Climate effects were near zero in *A. hyemalis* and *E. virginicus* (Pr($>$0) $\approx$ 0.46–0.49;Table S-8), but moderately positive in *P. autumnalis* (Pr = 0.87). Herbivory tended to reduce growth in the first two species (Pr $\approx$ 0.10–0.17) while increasing it in *P. autumnalis* (Pr = 0.93). The three-way interaction was only strongly negative in *P. autumnalis* (mean = –0.55; Pr $<$ 0.01;Table S-8). Reproduction was enhanced under combined endophyte and climate or herbivory effects. Flowering and spikelet output increased with Endophyte × Climate (IRR = 1.09–1.18; Pr(IRR $>$ 1) $>$ 0.66;Table S-8) and Endophyte × Herbivory in *P. autumnalis* (IRR = 1.31–1.96; Pr $>$ 0.96;Table S-8), whereas three-way interactions were negligible.

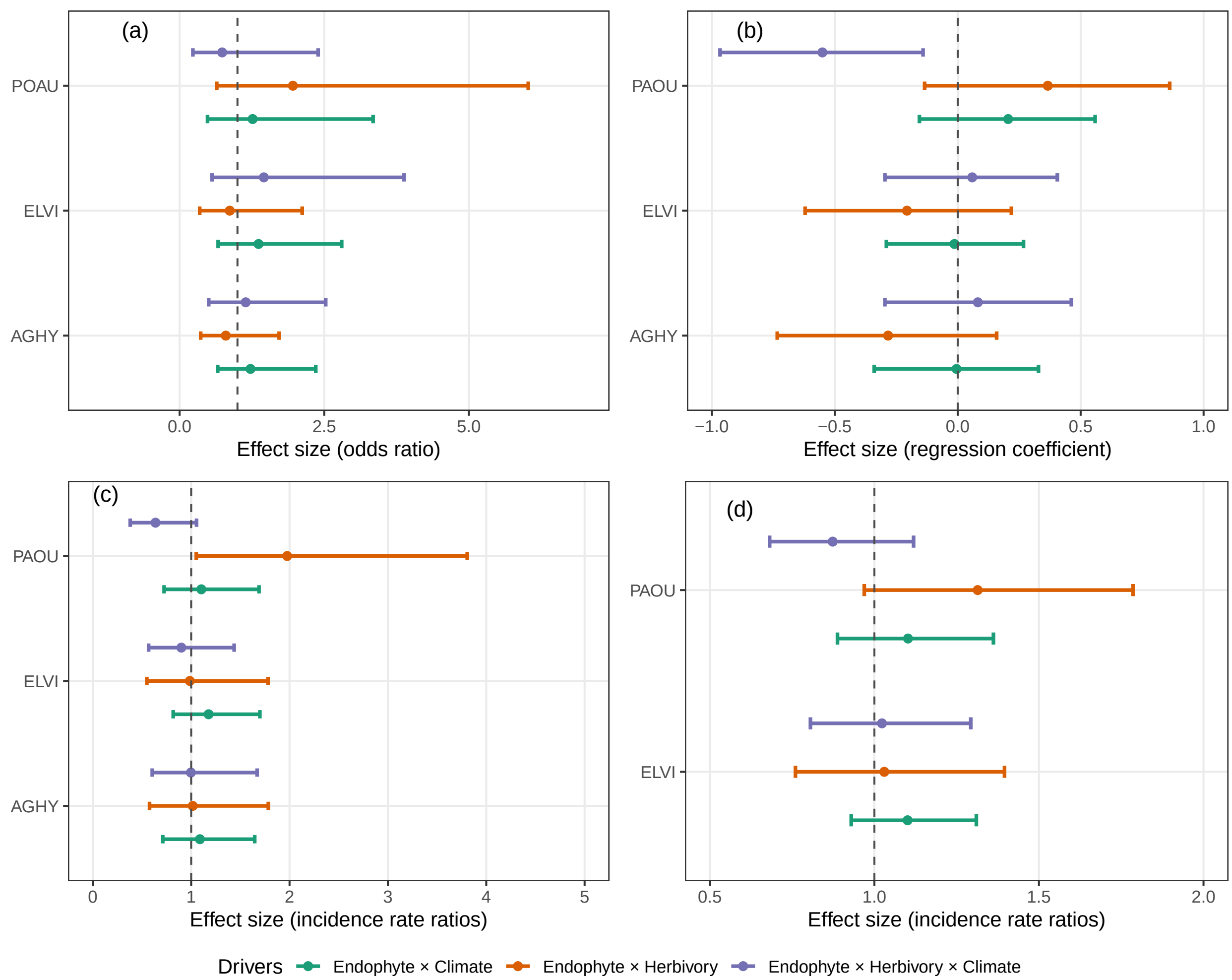

Figure 6: Standardized coefficients from Bayesian models showing how endophytes interact with climate and herbivore access to influence demographic rates of host grasses. Panels correspond to (a) survival, (b) growth, (c) flowering (inflorescence production), and (d) spikelet production. Coefficients represent standardized effect sizes for interaction terms: Endophyte × Climate, Endophyte × Herbivory, and Endophyte × Herbivory × Climate for the three species shown in the plot: *Agrostis hyemalis* (AGHY), *Elymus virginicus* (ELVI), and *Poa autumnalis* (POAU).

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Supporting Information

S.0.1 Candidate models for abiotic and biotic drivers of cool-grass species

1. **Abiotic model:** includes precipitation (P) and its quadratic term (P^2) as predictors, along with endophyte status (endo) and its interactions with climate. Species-specific effects borrow strength across species:

$$\mu_i = \beta_0[\text{Sp}_i] + \beta_{\text{endo}}[\text{Sp}_i]endo_i + \beta_P[\text{Sp}_i]P_i + \beta_{P^2}[\text{Sp}_i]P_i^2 + \beta_{\text{endo}\times P}[\text{Sp}_i]endo_i P_i + \beta_{\text{endo}\times P^2}[\text{Sp}_i]endo_i P_i^2$$
$$+ \phi_{\text{site}[i],\text{Sp}_i} + \omega_{\text{source}[i],\text{Sp}_i} + \rho_{\text{plot}[i],\text{Sp}_i}$$

2. **Biotic model:** includes endophyte status (endo), herbivory (herb), and their interaction, with species-specific effects:

$$\mu_i = \beta_0[\text{Sp}_i] + \beta_{\text{endo}}[\text{Sp}_i]endo_i + \beta_{\text{herb}}[\text{Sp}_i]herb_i + \beta_{\text{endo}\times\text{herb}}[\text{Sp}_i]endo_i herb_i$$
$$+ \phi_{\text{site}[i],\text{Sp}_i} + \omega_{\text{source}[i],\text{Sp}_i} + \rho_{\text{plot}[i],\text{Sp}_i}$$

3. **Full model:** combines abiotic and biotic predictors, including all relevant interactions up to the three-way interaction between endophyte, herbivory, and precipitation, with species-varying coefficients:

$$\mu_i = \beta_0[\text{Sp}_i] + \beta_P[\text{Sp}_i]P_i + \beta_{P^2}[\text{Sp}_i]P_i^2 + \beta_{\text{endo}}[\text{Sp}_i]endo_i + \beta_{\text{herb}}[\text{Sp}_i]herb_i$$
$$+ \beta_{\text{endo}\times P}[\text{Sp}_i]endo_i P_i + \beta_{\text{endo}\times P^2}[\text{Sp}_i]endo_i P_i^2 + \beta_{\text{herb}\times P}[\text{Sp}_i]herb_i P_i + \beta_{\text{herb}\times\text{endo}}[\text{Sp}_i]herb_i endo_i$$
$$+ \beta_{\text{endo}\times\text{herb}\times P}[\text{Sp}_i]endo_i herb_i P_i + \phi_{\text{site}[i],\text{Sp}_i} + \omega_{\text{source}[i],\text{Sp}_i} + \rho_{\text{plot}[i],\text{Sp}_i}$$

In these models, the β coefficients represent species-specific effects. The intercept $\beta_0[\text{Sp}_i]$ corresponds to the baseline value of the vital rate for species Sp_i in the absence of predictors. $\beta_{\text{endo}}[\text{Sp}_i]$ and $\beta_{\text{herb}}[\text{Sp}_i]$ describe the effects of endophyte presence and herbivory, respectively, for species Sp_i. The linear and quadratic effects of precipitation are captured by $\beta_P[\text{Sp}_i]$ and $\beta_{P^2}[\text{Sp}_i]$, while interactions between endophyte and precipitation are represented by $\beta_{\text{endo}\times P}[\text{Sp}_i]$ and $\beta_{\text{endo}\times P^2}[\text{Sp}_i]$. The interaction between endophyte and herbivory is given by $\beta_{\text{endo}\times\text{herb}}[\text{Sp}_i]$, the interaction between herbivory and precipitation by $\beta_{\text{herb}\times P}[\text{Sp}_i]$, and the three-way interaction among endophyte, herbivory, and precipitation by $\beta_{\text{endo}\times\text{herb}\times P}[\text{Sp}_i]$. All β coefficients are modeled hierarchically across species, allowing information to be shared while capturing species-specific responses. All models included a hierarchical random effects structure accounting for site-year, source population, and plot variation, with species-specific random effects $\phi_{\text{site}[i],\text{Sp}_i} \sim \mathcal{N}(0,\sigma_{\text{site}})$, $\omega_{\text{source}[i],\text{Sp}_i} \sim \mathcal{N}(0,\sigma_{\text{source}})$, and $\rho_{\text{plot}[i],\text{Sp}_i} \sim \mathcal{N}(0,\sigma_{\text{plot}})$.

S.1 Supporting Tables

Table S-1: Geographic coordinates and brief descriptions of source populations for *Agrostis hyemalis* (AGHY), *Elymus virginicus* (ELVI), and *Poa autumnalis* (POAU).

Species	Population	Latitude	Longitude	Description
AGHY	SJC	30.771969	-95.200620	Ecolab property
AGHY	COHR	29.672833	-96.567517	Columbus, TX herbarium record
AGHY	LPEF	30.085383	-97.172508	Lost Pines
POAU	LA7	30.849660	-93.276772	DeRidder, LA
POAU	LA1	32.568125	-93.932976	Jean Lafitte National Historical Park and Preserve
POAU	LA2	31.183040	-92.616969	Jean Lafitte National Historical Park and Preserve
POAU	SFAEF	31.492158	-94.772200	Stephen F. Austin Experimental Forest, TX
POAU	WB	30.399346	-95.150026	Sam Houston National Forest
ELVI	HUNT	30.741797	-95.474508	Sam Houston State University Field Station
ELVI	SHS	30.741797	-95.474508	Sam Houston State University Field Station
ELVI	RL5	30.471717	-99.783803	Texas Tech University, Junction
ELVI	PALM	29.591623	-97.584781	Palmetto State Park, TX
ELVI	JLP	29.785105	-90.114933	Jean Lafitte Park, outside of New Orleans

Table S-2: Common garden and field experiment sites used for reciprocal transplant and environmental manipulation studies.

Site	**Code**	**Latitude**	**Longitude**
University of Louisiana Ecology Center	LAF	30.304773	-92.012401
Center for Biological Field Studies	HUN	30.742815	-95.476881
TAMU Ecology and Natural Resource Teaching Area	COL	30.569302	-96.366324
UT Stengl Lost Pines Field Station	BAS	30.092343	-97.172442
1335 Medina Hwy, Kerrville, TX	KER	30.028437	-99.142888
Brackenridge Field Laboratory	BFL	30.284866	-97.780675
TAMU Sonora Station	SON	30.273467	-100.561463

Table S-3: Comparison of candidate models (quadratic vs. monotonic) for each vital rate based on leave-one-out cross-validation (LOOCV). ELPD difference is relative to the best model.

Vital rate	Model type	ELPD difference	SE difference
Survival	Quadratic	0	0.000
Survival	Monotonic	-0.653	0.997
Growth	Quadratic	0	0.000
Growth	Monotonic	-0.310	1.040
Flowering	Quadratic	0	0.000
Flowering	Monotonic	-0.562	1.010
Spikelet	Quadratic	0	0.078
Spikelet	Monotonic	-0.078	1.300

Table S-4: Posterior summaries of Δ ($E^{+} - E^{-}$) for survival at precipitation quantiles for each species and herbivore exclusion treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
Agrostis hyemalis	Unfenced	765.76	-0.0126	-0.1169	0.0719	0.3685
Agrostis hyemalis	Unfenced	848.28	-0.0099	-0.1011	0.0686	0.3965
Agrostis hyemalis	Unfenced	1113.25	0.0017	-0.0718	0.0759	0.5192
Agrostis hyemalis	Unfenced	1644.36	0.0258	-0.0675	0.1472	0.6943
Agrostis hyemalis	Unfenced	2163.28	0.0411	-0.0870	0.2249	0.7290
Agrostis hyemalis	Fenced	765.76	-0.0508	-0.1728	0.0156	0.0998
Agrostis hyemalis	Fenced	848.28	-0.0467	-0.1532	0.0189	0.1105
Agrostis hyemalis	Fenced	1113.25	-0.0278	-0.1131	0.0376	0.2291
Agrostis hyemalis	Fenced	1644.36	0.0134	-0.0899	0.1304	0.6016
Agrostis hyemalis	Fenced	2163.28	0.0439	-0.0888	0.2178	0.7282
Elymus virginicus	Unfenced	765.76	-0.0285	-0.1885	0.1340	0.3746
Elymus virginicus	Unfenced	848.28	-0.0174	-0.1651	0.1345	0.4170
Elymus virginicus	Unfenced	1113.25	0.0195	-0.1028	0.1465	0.6085
Elymus virginicus	Unfenced	1644.36	0.0765	-0.0642	0.2293	0.8173
Elymus virginicus	Unfenced	2163.28	0.1053	-0.0723	0.3098	0.8399
Elymus virginicus	Fenced	765.76	-0.1265	-0.2861	0.0322	0.0948
Elymus virginicus	Fenced	848.28	-0.0934	-0.2375	0.0457	0.1362
Elymus virginicus	Fenced	1113.25	-0.0019	-0.1130	0.1083	0.4883
Elymus virginicus	Fenced	1644.36	0.0883	-0.0246	0.2462	0.9023
Elymus virginicus	Fenced	2163.28	0.1131	-0.0047	0.3343	0.9435
Poa autumnalis	Unfenced	765.76	-0.0133	-0.1064	0.0219	0.1818
Poa autumnalis	Unfenced	848.28	-0.0226	-0.1354	0.0263	0.1708
Poa autumnalis	Unfenced	1113.25	0.0195	-0.1028	0.1465	0.1458
Poa autumnalis	Unfenced	1644.36	0.0769	-0.1765	0.0769	0.2885
Poa autumnalis	Unfenced	2163.28	0.0719	-0.1142	0.4382	0.4382
Poa autumnalis	Fenced	765.76	0.0060	-0.0338	0.0885	0.6755
Poa autumnalis	Fenced	848.28	0.0104	-0.0435	0.1079	0.6883
Poa autumnalis	Fenced	1113.25	0.0315	-0.0721	0.1574	0.7088
Poa autumnalis	Fenced	1644.36	0.0204	-0.0984	0.1579	0.6371
Poa autumnalis	Fenced	2163.28	0.0030	-0.0898	0.1139	0.5673

Table S-5: Posterior summaries of Δ ($E^{+} - E^{-}$) for growth across precipitation quantiles for each species and herbivore treatment. Positive median values indicate beneficial effects of endophytes; negative medians indicate costly effects.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
Agrostis hyemalis	Unfenced	773.91	-0.0058	-0.5523	0.4940	0.5608
Agrostis hyemalis	Unfenced	1075.09	0.0555	-0.3212	0.5970	0.6812
Agrostis hyemalis	Unfenced	1316.25	0.0929	-0.2332	0.4248	0.7393
Agrostis hyemalis	Unfenced	1979.35	0.1692	-0.2486	0.5977	0.7744
Agrostis hyemalis	Unfenced	2163.28	0.1868	-0.2787	0.6595	0.7724
Agrostis hyemalis	Fenced	773.91	0.0129	-0.4028	0.5208	0.6535
Agrostis hyemalis	Fenced	1075.09	0.0727	-0.2255	0.3735	0.6589
Agrostis hyemalis	Fenced	1316.25	0.1082	-0.1698	0.3944	0.7393
Agrostis hyemalis	Fenced	1979.35	0.1809	-0.2083	0.5777	0.7744
Agrostis hyemalis	Fenced	2163.28	0.1975	-0.2318	0.6393	0.7724
Elymus virginicus	Unfenced	773.91	-0.1229	-0.6743	0.3539	0.3539
Elymus virginicus	Unfenced	1075.09	-0.0079	-0.4102	0.4873	0.4873
Elymus virginicus	Unfenced	1316.25	0.0626	-0.2815	0.6218	0.6218
Elymus virginicus	Unfenced	1979.35	0.2071	-0.1825	0.8061	0.8061
Elymus virginicus	Unfenced	2163.28	0.2379	-0.1845	0.8208	0.8208
Elymus virginicus	Fenced	773.91	-0.0684	-0.5652	0.4161	0.4161
Elymus virginicus	Fenced	1075.09	-0.0264	-0.3481	0.4470	0.4470
Elymus virginicus	Fenced	1316.25	-0.0004	-0.2469	0.4988	0.4988
Elymus virginicus	Fenced	1979.35	0.0524	-0.2399	0.6128	0.6128
Elymus virginicus	Fenced	2163.28	0.0625	-0.2673	0.6203	0.6203
Poa autumnalis	Unfenced	773.91	0.1916	-0.4565	0.6861	0.6861
Poa autumnalis	Unfenced	1075.09	0.2583	-0.1590	0.8455	0.8455
Poa autumnalis	Unfenced	1316.25	0.3025	0.0011	0.9508	0.9508
Poa autumnalis	Unfenced	1979.35	0.3888	0.1279	0.9929	0.9929
Poa autumnalis	Unfenced	2163.28	0.4076	0.1048	0.9875	0.9875
Poa autumnalis	Fenced	773.91	1.1794	0.5397	1.8184	0.9989
Poa autumnalis	Fenced	1075.09	0.9284	0.5167	1.3384	1.0000
Poa autumnalis	Fenced	1316.25	0.7718	0.4755	1.0747	1.0000
Poa autumnalis	Fenced	1979.35	0.4611	0.1732	0.7596	0.9958
Poa autumnalis	Fenced	2163.28	0.3940	0.0647	0.7341	0.9773

Table S-6: Posterior summaries of Δ ($E^+ - E^-$) for flowering output at precipitation quantiles for each species and herbivore exclusion treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	P(Δ > 0)
Agrostis hyemalis	Unfenced	773.91	-0.0073	-0.75299	0.86870	0.4940
Agrostis hyemalis	Unfenced	1075.09	0.1156	-0.72258	1.03730	0.5970
Agrostis hyemalis	Unfenced	1316.25	0.2627	-0.72466	1.37065	0.6812
Agrostis hyemalis	Unfenced	1979.35	0.8796	-1.36442	4.4190	0.7445
Agrostis hyemalis	Unfenced	2163.28	1.0972	-1.75172	5.9933	0.7410
Agrostis hyemalis	Fenced	773.91	0.0223	-0.76233	0.93648	0.5208
Agrostis hyemalis	Fenced	1075.09	0.2099	-0.69651	1.25368	0.6589
Agrostis hyemalis	Fenced	1316.25	0.4266	-0.70743	1.80962	0.7393
Agrostis hyemalis	Fenced	1979.35	1.3034	-1.61430	5.89453	0.7744
Agrostis hyemalis	Fenced	2163.28	1.6062	-2.06481	7.75656	0.7724
Elymus virginicus	Unfenced	773.91	-0.0709	-0.44121	0.28281	0.3539
Elymus virginicus	Unfenced	1075.09	-0.0056	-0.34000	0.31918	0.4873
Elymus virginicus	Unfenced	1316.25	0.0542	-0.27674	0.39898	0.6218
Elymus virginicus	Unfenced	1979.35	0.2372	-0.22767	0.89597	0.8061
Elymus virginicus	Unfenced	2163.28	0.2921	-0.24417	1.10392	0.8208
Elymus virginicus	Fenced	773.91	-0.0964	-0.87499	0.79622	0.4161
Elymus virginicus	Fenced	1075.09	-0.0476	-0.66448	0.62745	0.4470
Elymus virginicus	Fenced	1316.25	-0.0006	-0.55277	0.58266	0.4988
Elymus virginicus	Fenced	1979.35	0.1347	-0.67562	1.11263	0.6128
Elymus virginicus	Fenced	2163.28	0.1640	-0.79414	1.37908	0.6203
Poa autumnalis	Unfenced	773.91	0.0029	-0.02154	0.07747	0.6861
Poa autumnalis	Unfenced	1075.09	0.0414	-0.02992	0.29571	0.8455
Poa autumnalis	Unfenced	1316.25	0.1812	0.00072	0.82475	0.9508
Poa autumnalis	Unfenced	1979.35	2.8401	0.31057	20.2205	0.9929
Poa autumnalis	Unfenced	2163.28	5.0522	0.37729	46.2966	0.9875
Poa autumnalis	Fenced	773.91	0.0744	0.00658	0.64876	0.9989
Poa autumnalis	Fenced	1075.09	0.3639	0.07496	1.53393	1.0000
Poa autumnalis	Fenced	1316.25	0.9483	0.23684	3.28046	1.0000
Poa autumnalis	Fenced	1979.35	5.2444	0.61659	36.1808	0.9958
Poa autumnalis	Fenced	2163.28	6.8435	0.35587	65.0251	0.9773

Table S-7: Posterior summaries of Δ ($E^+ - E^-$) at precipitation quantiles for each species and herbivore exclusion treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
Elymus virginicus	Unfenced	1035.41	-1.4080	-5.6588	2.5136	0.272
Elymus virginicus	Unfenced	1276.99	-0.6715	-4.3261	2.8829	0.371
Elymus virginicus	Unfenced	1644.36	0.3298	-3.1403	3.9781	0.564
Elymus virginicus	Unfenced	2163.28	1.5276	-2.5967	6.4739	0.737
Elymus virginicus	Unfenced	2182.71	1.5725	-2.5800	6.5641	0.741
Elymus virginicus	Fenced	1035.41	-1.0142	-4.4019	2.6090	0.306
Elymus virginicus	Fenced	1276.99	0.0749	-2.6481	2.9073	0.517
Elymus virginicus	Fenced	1644.36	1.5186	-1.0554	4.5225	0.839
Elymus virginicus	Fenced	2163.28	3.2830	-0.5052	8.7059	0.922
Elymus virginicus	Fenced	2182.71	3.3381	-0.4955	8.9005	0.923
Poa autumnalis	Unfenced	1035.41	-1.2269	-5.1615	2.0682	0.253
Poa autumnalis	Unfenced	1276.99	-0.7580	-3.6361	1.9503	0.314
Poa autumnalis	Unfenced	1644.36	0.1225	-2.2064	2.5035	0.538
Poa autumnalis	Unfenced	2163.28	1.4394	-2.0529	5.7896	0.764
Poa autumnalis	Unfenced	2182.71	1.4904	-2.0767	5.9711	0.766
Poa autumnalis	Fenced	1035.41	3.3952	-0.5175	8.5092	0.927
Poa autumnalis	Fenced	1276.99	3.5317	0.4555	7.5555	0.969
Poa autumnalis	Fenced	1644.36	3.5791	0.7899	7.3800	0.984
Poa autumnalis	Fenced	2163.28	3.4620	-0.8494	9.3572	0.910
Poa autumnalis	Fenced	2182.71	3.4513	-0.9420	9.4863	0.906

Table S-8: Posterior summaries of interaction effects (mean estimate, 95% credible intervals, and posterior probability) for survival, growth, flowering, and spikelet production across three grass species.

Fitness component	Species	Parameter	Estimate	Lower CI	Upper CI	Probability
Survival	*Agrostis hyemalis*	Endophyte × Climate	1.213	0.658	2.354	0.730
Survival	*Elymus virginicus*	Endophyte × Climate	1.364	0.666	2.802	0.806
Survival	*Poa autumnalis*	Endophyte × Climate	1.255	0.483	3.344	0.681
Survival	*Agrostis hyemalis*	Endophyte × Herbivory	0.804	0.364	1.720	0.287
Survival	*Elymus virginicus*	Endophyte × Herbivory	0.868	0.347	2.119	0.376
Survival	*Poa autumnalis*	Endophyte × Herbivory	1.941	0.642	6.026	0.879
Survival	*Agrostis hyemalis*	Endophyte × Herbivory × Climate	1.149	0.503	2.525	0.630
Survival	*Elymus virginicus*	Endophyte × Herbivory × Climate	1.450	0.560	3.880	0.774
Survival	*Poa autumnalis*	Endophyte × Herbivory × Climate	0.737	0.229	2.394	0.303
Growth	*Agrostis hyemalis*	Endophyte × Climate	-0.005	-0.340	0.329	0.488
Growth	*Elymus virginicus*	Endophyte × Climate	-0.013	-0.290	0.268	0.461
Growth	*Poa autumnalis*	Endophyte × Climate	0.208	-0.156	0.559	0.873
Growth	*Agrostis hyemalis*	Endophyte × Herbivory	-0.283	-0.734	0.158	0.107
Growth	*Elymus virginicus*	Endophyte × Herbivory	-0.208	-0.621	0.218	0.169
Growth	*Poa autumnalis*	Endophyte × Herbivory	0.365	-0.135	0.862	0.929
Growth	*Agrostis hyemalis*	Endophyte × Herbivory × Climate	0.084	-0.296	0.462	0.666
Growth	*Elymus virginicus*	Endophyte × Herbivory × Climate	0.060	-0.296	0.405	0.634
Growth	*Poa autumnalis*	Endophyte × Herbivory × Climate	-0.549	-0.966	-0.141	0.004
Flowering	*Agrostis hyemalis*	bendoclim	1.092	0.711	1.646	0.656
Flowering	*Elymus virginicus*	bendoclim	1.178	0.817	1.698	0.805
Flowering	*Poa autumnalis*	bendoclim	1.103	0.725	1.689	0.676
Flowering	*Agrostis hyemalis*	bendoherb	1.014	0.577	1.784	0.523
Flowering	*Elymus virginicus*	bendoherb	0.987	0.549	1.781	0.478
Flowering	*Poa autumnalis*	bendoherb	1.964	1.052	3.807	0.982
Flowering	*Agrostis hyemalis*	bendoherbclim	0.997	0.604	1.670	0.496
Flowering	*Elymus virginicus*	bendoherbclim	0.900	0.567	1.437	0.332
Flowering	*Poa autumnalis*	bendoherbclim	0.640	0.380	1.055	0.038
Spikelet	*Elymus virginicus*	bendoclim	1.100	0.930	1.310	0.866
Spikelet	*Poa autumnalis*	bendoclim	1.103	0.888	1.362	0.817
Spikelet	*Elymus virginicus*	bendoherb	1.029	0.760	1.395	0.573
Spikelet	*Poa autumnalis*	bendoherb	1.315	0.969	1.786	0.961
Spikelet	*Elymus virginicus*	bendoherbclim	1.023	0.806	1.293	0.573
Spikelet	*Poa autumnalis*	bendoherbclim	0.873	0.682	1.119	0.145

S.2 Supporting Figures

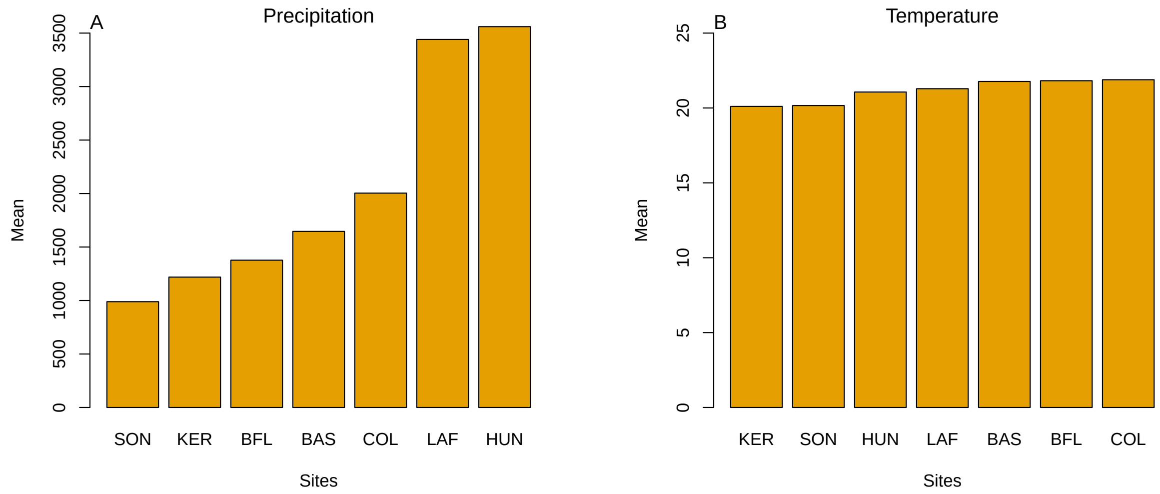

Figure S-1: Abiotic conditions across the seven study sites. (A) Mean annual precipitation showing the pronounced east–west aridity gradient from mesic Louisiana to arid Texas. (B) Mean annual temperature across sites, which does not differ much. Panels highlight that precipitation, rather than temperature, constitutes the primary abiotic gradient in this system.

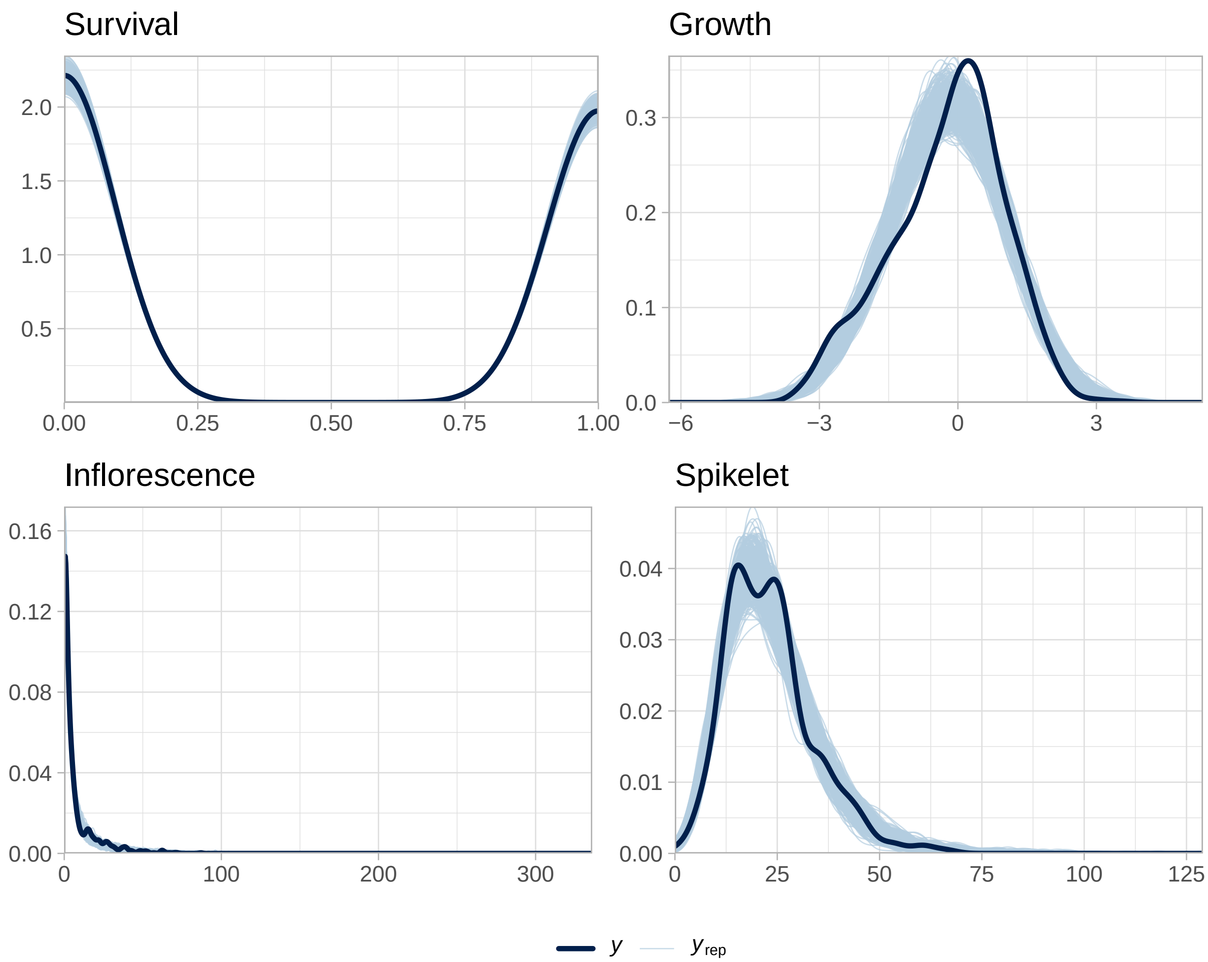

Figure S-2: Posterior predictive checks for the full demographic model. Dark blue lines show observed data (y), and light blue lines show replicated datasets generated from posterior predictions (y_{rep}). The strong overlap across survival, growth, flowering, and spikelet production indicates adequate model fit for all vital rates.

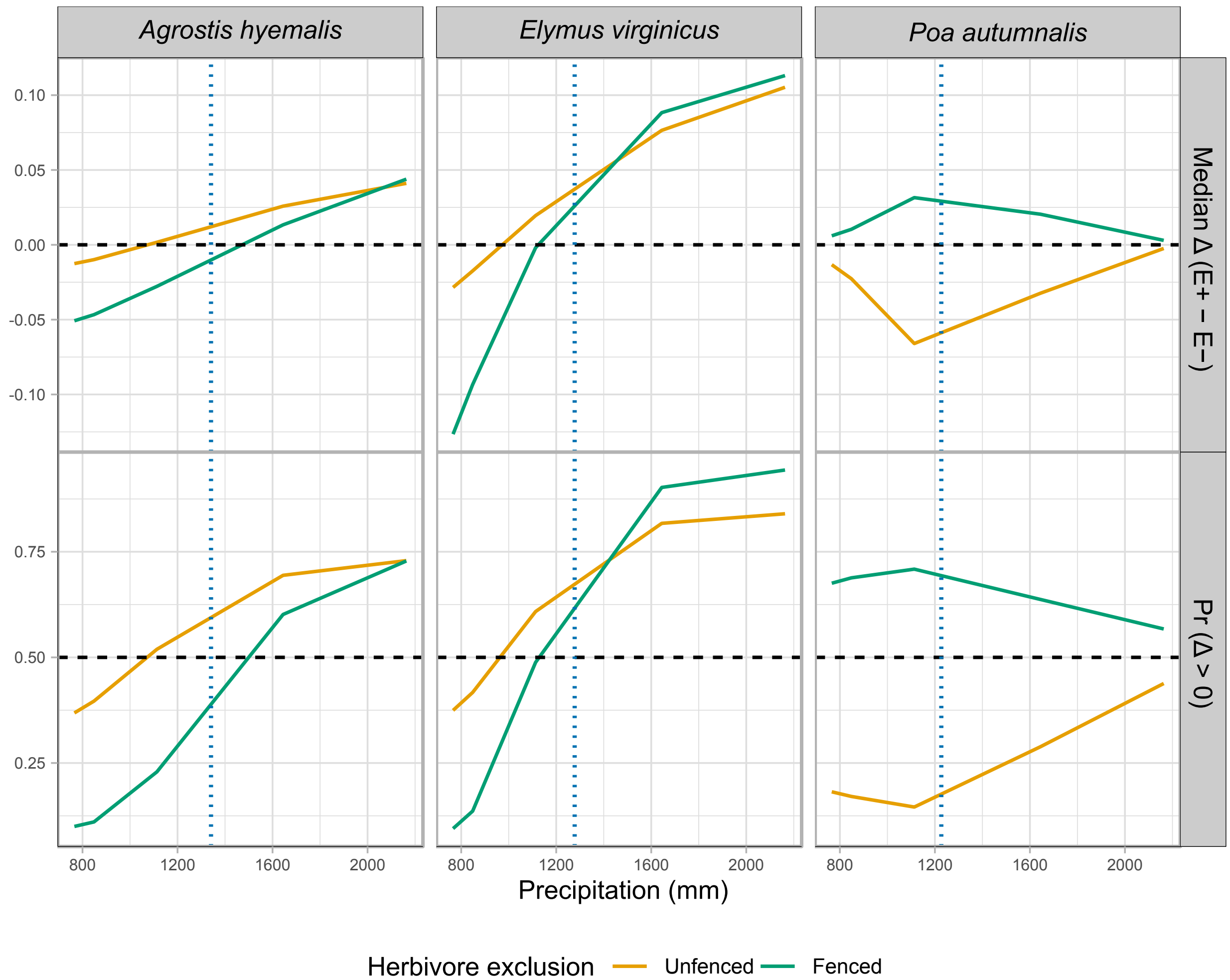

Figure S-3: Predicted effects of endophyte presence (E^+) versus absence (E^-) on survival across precipitation gradients for three cool-season grasses. Solid lines show the median Δ ($E^+ - E^-$),or the posterior probability that $\Delta > 0$ under fenced and unfenced conditions. Horizontal dashed lines indicate no effect ($\Delta = 0$) or a 50% probability of benefit. Vertical dashed lines represent the precipitation values at the most westerly natural occurrence of each species, based on GBIF records.

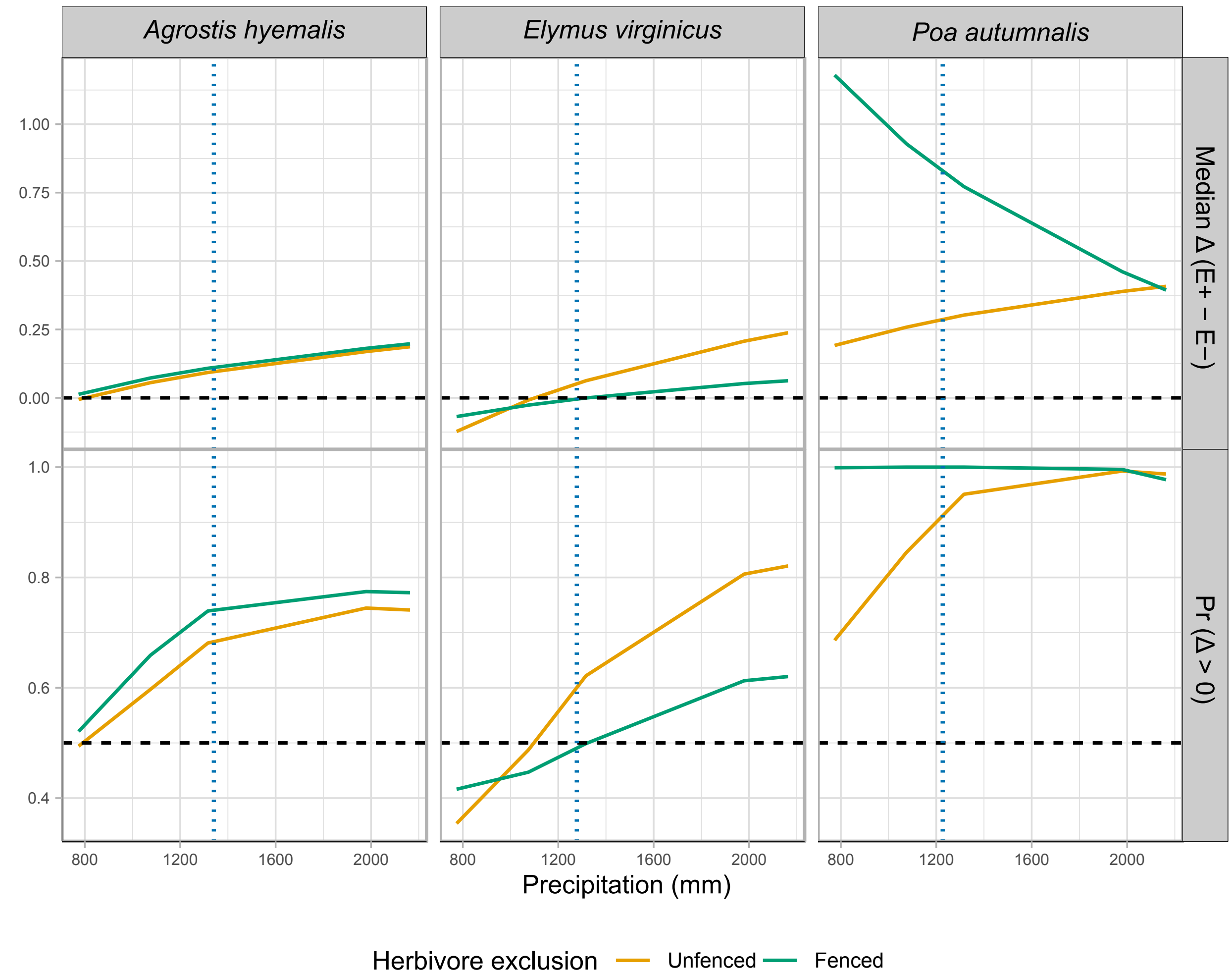

Figure S-4: Effects of precipitation and endophyte presence on growth across three cool-season grass species. Lines show the predicted difference in growth between endophyte-present (E^+) and endophyte-absent (E^-) plants (Median Δ, top row) and the posterior probability that this difference is positive ($\Pr(\Delta > 0)$, bottom row) under fenced and unfenced treatments. Predictions were calculated at the 10th, 25th, 50th, 75th, and 90th percentiles of precipitation observed in the study. Horizontal dashed lines indicate $\Delta = 0$ (top row) and $\Pr(\Delta > 0) = 0.5$ (bottom row). Vertical dashed lines represent the precipitation values at the most westerly natural occurrence of each species, based on GBIF records.

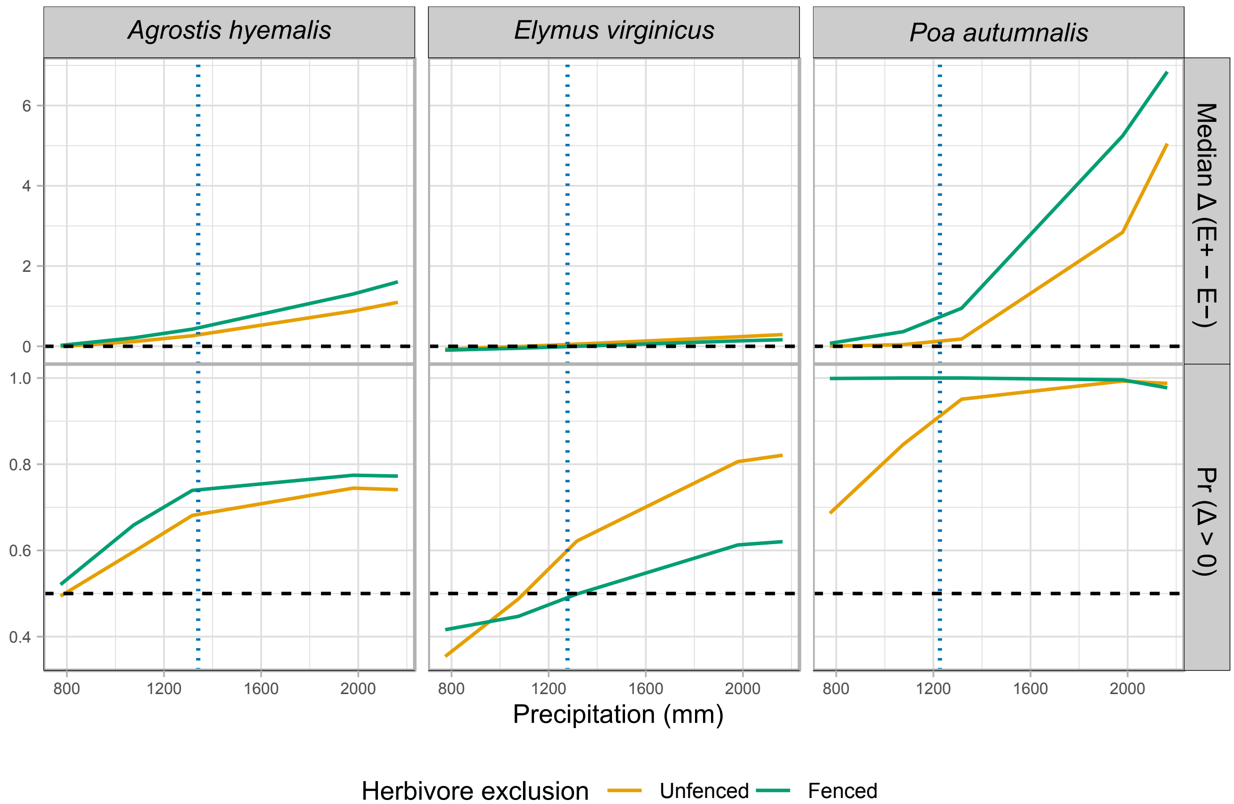

Figure S-5: Effects of endophyte infection on flowering output across a precipitation gradient for three cool-season grass species. Lines show the median posterior difference Δ ($E^+ - E^-$) and the posterior probability that $\Delta > 0$ under fenced (herbivores excluded) and unfenced (herbivores present) conditions. Horizontal dashed lines indicate $\Delta = 0$ for median differences and $\Pr(\Delta > 0) = 0.5$. Endophyte effects were generally positive for *Agrostis hyemalis* and *Poa autumnalis*, but for *Elymus virginicus* benefits were context-dependent, with costs at lower precipitation

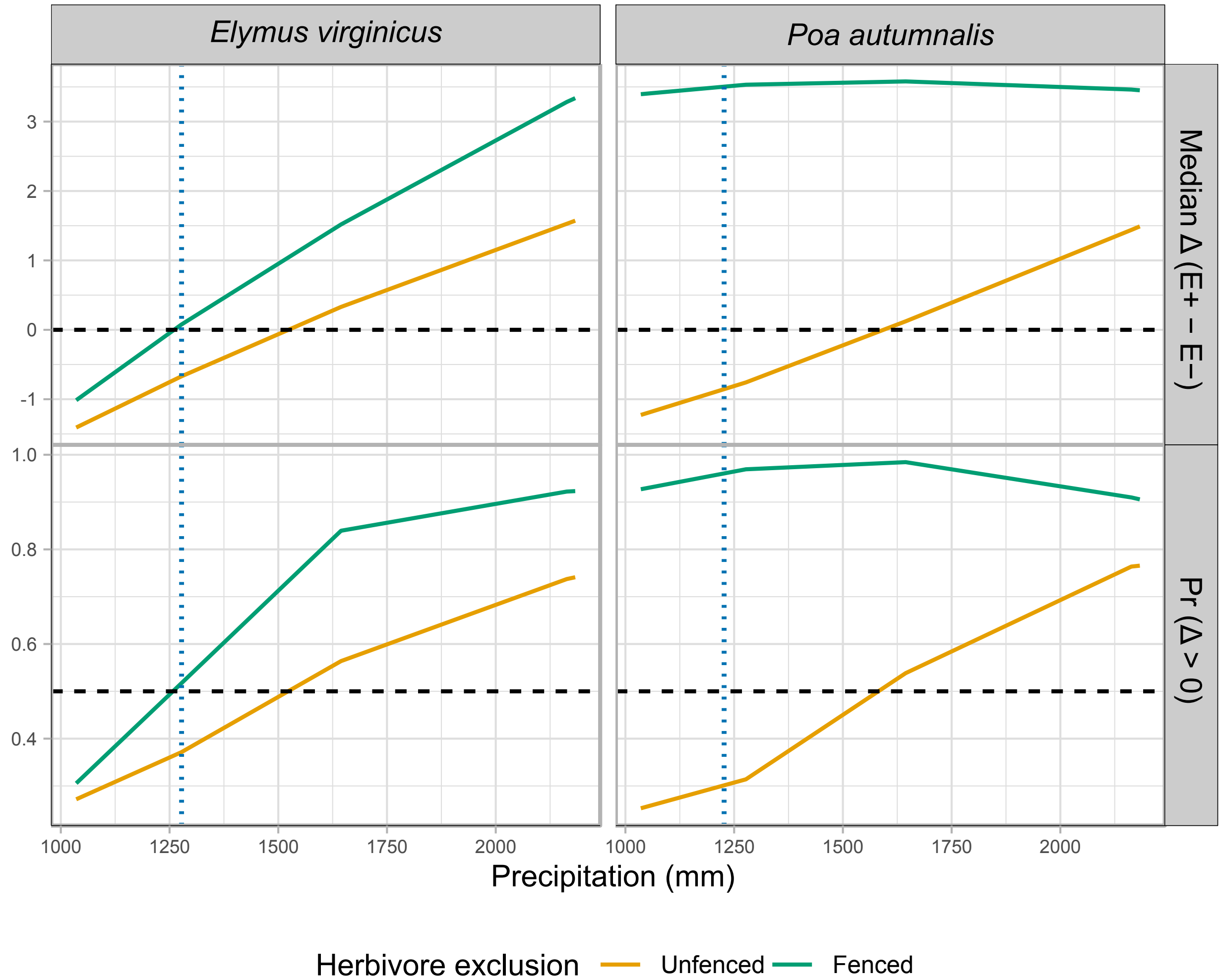

Figure S-6: Effects of precipitation and endophyte presence on spike (reproductive output) for three cool-season grass species. Median Δ ($E^+ - E^-$) and posterior probability $\Pr(\Delta > 0)$ are shown across precipitation quantiles. Herbivore exclusion treatments are indicated by color (Fenced = herbivores excluded, Unfenced = herbivores had access). Horizontal dashed lines indicate $\Delta = 0$ for medians and $\Pr(\Delta > 0) = 0.5$ for probabilities. Facets separate metrics and species.